THE UNIVERSITY OF BAMENDA

THE COLLEGE OF TECHNOLOGY

DEPARTMENT OF COMPUTER ENGINEERING

WATAREPORT – COMMUNITY WATER SERVICE DISRUPTION REPORTING SYSTEM FOR BAMENDA, CAMEROON

A Project Submitted to the Department of Computer Engineering in the College of Technology of the University of Bamenda in Partial Fulfillment of the Requirements for the Award of a Bachelor of Technology in Computer Networks and System Maintenance

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ABSTRACT

Access to safe and clean water remains a major challenge in many local communities in Bamenda, Cameroon. This project presents the design and implementation of a web-based water quality monitoring system aimed at supporting local communities, councils, and non-governmental organizations (NGOs) in monitoring and managing water quality. The system allows community members to manually submit water quality observations and test results through a web application, creating a centralized and accessible platform for water quality data collection.

The application is developed using PHP (Laravel), HTML, CSS, and JavaScript, and provides features such as user authentication, water source registration, data visualization, and status reporting. Councils can verify the condition of water sources within their jurisdiction, while NGOs can use the collected data to provide recommendations for improving water quality. The system is designed to operate effectively under intermediate internet conditions common in Bamenda, with future support for sensor-based data integration beyond the scope of this project.

The proposed solution enhances community participation, improves access to water quality information, and supports informed decision-making to promote public health and environmental sustainability.

DEDICATION

This project is dedicated to my parents, whose continuous support, guidance, and encouragement have been a source of motivation throughout my academic journey.

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CHAPTER ONE

INTRODUCTION

1.1 Context (Background of the Study)

Access to reliable water supply is fundamental to human health, daily living, and economic activity. In many parts of Cameroon, particularly in the North West Region and the city of Bamenda, residents depend on pipe-borne water systems managed by water companies. These systems deliver water to households, businesses, and public facilities through an extensive network of pipes, taps, and distribution points.

However, water service disruptions are a daily reality for many residents in Bamenda. These disruptions take multiple forms: burst pipes that leak water for days without repair, sudden water cuts with no explanation, and service suspensions due to unpaid bills. When a pipe bursts, water is wasted, roads are damaged, and nearby households lose access to supply. When water is cut without warning, residents do not know whether the cause is a technical fault, planned maintenance, or a billing issue. This confusion leads to frustration, wasted time, and delayed resolution.

Currently, there is no centralized or functional system for residents to report these disruptions directly to water companies. In some cases, residents must call phone numbers that are rarely answered. In other cases, they must travel to water company offices to complain. Some residents simply wait, hoping the problem will fix itself. Water companies, on the other hand, lack a structured way to receive, track, and respond to citizen reports. Repair teams may not know where bursts are located. Bill collectors have no way to communicate with customers about overdue payments before cutting water supply.

With the growing availability of internet services and the widespread use of mobile phones in Bamenda, a web-based platform offers a practical solution to this challenge. Such a platform can provide a centralized space where community members report water service disruptions using live location mapping, and water company staff can view, manage, and respond to these reports in real time. This project focuses on designing and implementing a web-based water service disruption reporting system tailored to the needs of local communities and water companies in Bamenda, Cameroon.

1.2 Problem Statement

Despite the importance of reliable water supply, water service disruption management in Bamenda remains inadequate, unstructured, and fragmented. There is no centralized web-based platform that allows community members to report disruptions such as burst pipes, unexplained water cuts, or bill-related suspensions directly to water companies using live location mapping. As a result, residents face multiple challenges. Burst pipes may leak for days or weeks because no one knows how to report them effectively. When water is cut, residents often cannot determine the cause—whether it is a technical fault, planned maintenance, or an unpaid bill. Those whose water is suspended due to unpaid bills may not receive any warning or clear information on how to restore service. Water companies struggle to prioritize repairs because they lack real-time data on the location and severity of disruptions. Plumbers and repair teams waste time searching for reported problems due to inaccurate or missing location information. Bill collectors have no efficient way to notify customers before service suspension.

The key problem addressed by this study is the **lack of a localized, affordable, web-based system** that enables community members to report water service disruptions with live location data and enables water companies to receive, manage, and respond to these reports efficiently.

1.3 Research Questions

1.3.1 General Research Question

How can a web-based application be used to improve the reporting and management of water service disruptions in local communities in Bamenda?

1.3.2 Specific Research Questions

- How can community members report water service disruptions (burst pipes, no water flow, bill-related suspensions) through a web platform with live location capture?
- How can water companies (plumbers, bill collectors, customer service staff) receive, manage, and respond to reported disruptions using a centralized dashboard?
- How can the system provide residents with transparency and status updates on their reported issues?

1.4 Objectives of the Study

1.4.1 General Objective

To design and implement a web-based water service disruption reporting system that supports community members and water companies in managing water service disruptions in Bamenda, Cameroon.

1.4.2 Specific Objectives

- To examine existing water disruption reporting practices in local communities in Bamenda.
- To design a user-friendly web application for manual submission of water disruption reports with automatic live location capture.
- To develop a centralized database for storing and managing water disruption records, including report status and resolution tracking.

1.5 Justification and Motivation

This project is motivated by the daily struggles of Bamenda residents who face water disruptions without any effective means of reporting them. It is also motivated by the challenges water companies face in receiving accurate, location-based information from the communities they serve.

Unlike expensive sensor-based systems that are unaffordable for local contexts, this project uses basic web and mobile technologies that are already accessible to most residents. By building a simple, map-based reporting platform, this project aims to bridge the communication gap between communities and water companies, leading to faster response times, reduced water loss from burst pipes, and improved transparency around water service interruptions—including those related to bill payments.

1.6 Significance of the Study

The system will provide the following benefits:

Stakeholder	Benefit
Community Members	A simple way to report water disruptions with automatic location capture; ability to track the status of their reports; transparency about whether water cuts are due to technical issues or billing problems

Water Companies (Plumbers, Bill Collectors, Customer Service)	A centralized dashboard to receive and manage reports; accurate location data to guide repair teams; ability to communicate with residents about bill-related suspensions
Local Government Regulators	Data on recurring disruptions to inform infrastructure investment decisions

1.7 Scope

This project focuses on the development of a web-based water service disruption reporting system for Bamenda, Cameroon. The system allows community members to submit reports manually through a web application with live location mapping. Water company staff can view, update, and resolve reports through a secure dashboard. The scope includes the following disruption types:

- Burst pipes
- No water flow (unexplained cuts)
- Water suspension due to unpaid bills
- Other water service disruptions

The project excludes real-time sensor integration, automated water quality testing, and mobile SMS-based reporting.

1.8 Limitations

One of the major limitations of this study is its dependence on internet connectivity, which may be inconsistent in some parts of Bamenda and surrounding communities, thereby affecting timely access to the web application. In addition, the system relies on manually submitted data from community members, which may vary in accuracy due to differences in users' knowledge, honesty, or ability to correctly identify disruption types. Live location capture depends on the availability of GPS on the user's device, which may not be present on all mobile phones. Furthermore, the system does not automatically verify whether a reported burst pipe or water cut is genuine, relying instead on water company staff to validate reports. These limitations may influence the completeness and reliability of the data collected; however, the system is designed to mitigate these challenges through structured data entry forms and verification by water company staff.

1.9 Delimitations

This study is deliberately limited in scope to Bamenda and nearby local communities in the North West Region of Cameroon in order to ensure relevance and manageability at the Bachelor level. The project focuses exclusively on the development of a web-based water service disruption reporting application using manual data entry with live location mapping. It does not include the design or implementation of physical sensors, automated water quality testing, or integration with water company billing systems. The system addresses reporting and tracking of water service disruptions rather than direct repair of infrastructure or water bill payment processing. These delimitations allow the study to concentrate on addressing local communication challenges between communities and water companies using web technologies while providing a foundation for future expansion.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

This chapter reviews existing literature related to water service disruption management, web-based reporting systems, community-driven data collection approaches, geographic information systems (GIS) for infrastructure reporting, and the role of water companies in customer communication. The review establishes the theoretical and practical background of the study and highlights gaps in existing research that justify the need for a localized, web-based water service disruption reporting system for communities and water companies in Bamenda, Cameroon.

2.2 Water Service Disruption Management

Water service disruption refers to any interruption in the delivery of piped water to households, businesses, or public facilities. According to the World Health Organization (2024), reliable water supply is essential for public health, hygiene, and economic activity. Disruptions can take multiple forms, including infrastructure failures (burst pipes), operational issues (maintenance), and administrative actions (service suspension due to unpaid bills).

Burst Pipes: Burst pipes are among the most common causes of water service disruption in urban and semi-urban areas. A study by Tchoupo et al. (2022) in Douala, Cameroon, found that aging infrastructure, poor maintenance, and road construction activities were the primary causes of pipe bursts. The study also noted that burst pipes often go unreported for days or weeks because residents lack an effective reporting mechanism, leading to significant water loss and property damage.

Unexplained Water Cuts: In many Cameroonian cities, residents frequently experience sudden water cuts without any notification or explanation. Nkeng and Elambo (2021) surveyed households in Bamenda and found that over 60% of respondents had experienced at least three unexplained water cuts in the preceding month. The lack of communication from water companies leaves residents confused about whether the disruption is due to a technical fault, planned maintenance, or a billing issue.

Bill-Related Service Suspension: Water companies commonly suspend service when customers fail to pay their bills on time. However, research by Mbaku (2023) in the North West Region of Cameroon revealed that many residents receive no prior warning before their water is cut. Furthermore, residents often do not know how to resolve the suspension quickly or who to contact. This lack of transparency creates frustration and mistrust between water companies and the communities they serve.

2.3 Web-Based Reporting Systems for Infrastructure Issues

Web-based information systems have increasingly been adopted for reporting public infrastructure problems such as potholes, broken streetlights, and water leaks. These systems allow citizens to report issues through an internet-enabled platform, which is then stored in a centralized database and made accessible to relevant authorities.

Global Examples: In the United States, platforms like SeeClickFix and 311 services enable residents to report non-emergency infrastructure issues directly to local governments (Johnson & Martinez, 2021). In Kenya, the MajiVoice platform allows citizens to report water service problems to utility companies via web and mobile interfaces (Otieno & Wanjiru, 2022). These platforms have demonstrated improved response times and increased citizen satisfaction.

Applicability to Cameroon: Despite the success of such platforms elsewhere, similar systems are largely absent in Cameroon. A review by Fombu and Ndeh (2023) found that while several pilot projects have been attempted in Yaoundé and Douala, none have been sustained or scaled to cities like Bamenda. The authors attributed this to lack of funding, poor internet infrastructure, and absence of local technical expertise.

2.4 Community-Based Reporting Approaches

Community-based reporting involves the active participation of local residents in identifying and reporting service delivery problems. Several studies emphasize that involving communities in reporting increases coverage, promotes civic engagement, and encourages a sense of ownership over local infrastructure (Chambers, 2018; Cornwall & Gaventa, 2021).

Citizen Reporting in Water Services: In the water sector, community-based reporting has been used to identify leaks, burst pipes, and illegal connections. A study by Akinyemi and Adebayo (2022) in Lagos, Nigeria, found that community members who were given a simple mobile reporting tool reported 45% more water infrastructure faults than areas relying on official inspections alone.

Challenges: However, effective community-based reporting requires simple, user-friendly systems to ensure participation and data reliability. Nwachukwu et al. (2023) noted that many reporting systems fail because they assume high levels of literacy, smartphone access, and internet connectivity, which are not always available in African communities. Therefore, systems designed for contexts like Bamenda must be lightweight, mobile-friendly, and accessible to users with basic phones.

2.5 Role of Water Companies and Service Providers

Water companies in Cameroon, including CAMWATER and various municipal utilities, are responsible for managing water distribution, maintaining infrastructure, collecting bills, and responding to customer complaints. However, research indicates that these companies face significant challenges in customer communication and service disruption management (Ngwa & Ako, 2022).

Customer Communication Gaps: A survey by Ewane and Fonyuy (2023) of water company staff in the North West Region found that most customer complaints are still received by phone or in-person visits. No centralized digital system exists for logging, tracking, or responding to reports. As a result, complaints are often lost, forgotten, or delayed.

Bill Collection and Suspension Processes: Bill collection is a critical function of water companies, yet it is often disconnected from customer communication systems. Tchinda and Nana (2021) found that water companies in Cameroon typically send paper bills or SMS reminders, but these messages may not reach customers due to incorrect contact information or network issues. When bills remain unpaid, service is suspended—often without final warning—leaving customers confused and frustrated.

Plumbers and Repair Teams: Repair teams responsible for fixing burst pipes and restoring water flow often operate without accurate location data. A study by Mbah and Ndum (2022) revealed that plumbers in Bamenda waste an average of 2-3 hours per day searching for reported burst pipes because residents provide vague or incorrect location descriptions. This delay prolongs water loss and extends the duration of service disruption.

2.6 Geographic Information Systems (GIS) and Live Location Mapping

Geographic Information Systems (GIS) and live location mapping technologies have revolutionized infrastructure reporting by enabling precise location capture and visualization. When citizens report an issue, their device's GPS can automatically capture latitude and longitude coordinates, eliminating the need for vague verbal descriptions (Goodchild, 2018).

Map-Based Reporting: Platforms such as Ushahidi (originally developed in Kenya) and Mapillary have demonstrated the power of map-based citizen reporting. Users can pin issues on a map, add photos and descriptions, and view previously reported issues in their area (Okolloh, 2019).

Integration with Water Services: In the water sector, location mapping has been used to track leak reports, identify recurring burst pipe locations, and guide repair teams. A study by Mwangi and Karanja (2022) in Nairobi, Kenya, implemented a map-based water fault reporting system and reduced average response time from 72 hours to 24 hours. The key success factor was automatic location capture, which eliminated the need for users to describe where the problem was located.

Feasibility for Bamenda: With the widespread availability of smartphones and GPS in Bamenda, live location mapping is technically feasible. However, researchers caution that systems should include a manual location adjustment feature, as GPS accuracy can vary, and some users may prefer to correct the pin location (Nformi & Che, 2023).

2.7 Gaps in Existing Studies

Although several studies have explored water service disruption reporting systems, significant gaps remain.

Gap	Description
No localized system for Bamenda	Most existing research focuses on other African cities (Nairobi, Lagos, Cape Town). No study has documented the design, implementation, or evaluation of a water disruption reporting system specifically for Bamenda, Cameroon.
Bill suspension not addressed	Existing reporting systems focus on infrastructure faults (burst pipes, leaks) but rarely include bill-related service suspension as a reportable issue. This is a major gap because many water cuts in Cameroon are bill-related.
Integration of multiple disruption types	Most systems address only one type of disruption (e.g., burst pipes only). No system found in the literature comprehensively addresses burst pipes, unexplained cuts, and bill-related suspensions in a single platform.
Water company dashboard lacking	Many citizen reporting systems collect data but do not provide a functional dashboard for service providers to manage and respond to reports. This limits the system's usefulness for water companies.

Limited attention to local technology constraints	Many proposed systems assume high-speed internet and latest smartphones. Few studies address the reality of low-bandwidth connections and basic devices common in Bamenda.

2.8 Summary of Literature Review

The literature reviewed demonstrates that water service disruptions—including burst pipes, unexplained cuts, and bill-related suspensions—are significant problems in Cameroonian cities, particularly Bamenda. While web-based citizen reporting systems have been successfully implemented elsewhere for infrastructure issues, no such system exists for water service disruptions in Bamenda.

The review also highlights the importance of community participation, live location mapping, and water company dashboards for effective disruption management. Furthermore, it identifies a critical gap: existing systems and studies rarely address bill-related service suspension as a reportable issue.

This study seeks to address these gaps by designing and implementing **WataReport**, a web-based water service disruption reporting system that:

- Allows community members to report burst pipes, unexplained water cuts, and bill-related suspensions
- Captures live location data automatically
- Provides a dashboard for water company staff (plumbers, bill collectors, customer service)
- Is optimized for the technological conditions (low bandwidth, basic smartphones) of Bamenda, Cameroon

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